Outline

1) Score function
2) Fisher information
3) Cramér-Rao Lower Bound
4) Examples

Score function

Assume $\mathcal{P}$ has densities p_θ wrt μ, $\Theta \subseteq \mathbb{R}^d$

Common support: $\{x : p_\theta(x) > 0\}$ same $\forall \theta$

Log-likelihood: $\ell(\theta; X) = \log p_\theta(X)$,

Thought of as random function of θ

$\ell(\cdot\,; X) - \ell(\theta_0; X)$ minimal suff for any fixed $\theta_0 \in \Theta$

Asymptotic picture: What if X highly informative?

Easy to rule out P_θ unless $\theta \approx \theta_0$ (true value)

$\Rightarrow$ "effective model" is $\{P_{\theta_0 + \eta} : \eta \approx 0\}$

$$\Rightarrow \underbrace{\ell(\theta_0 + \eta; X) - \ell(\theta_0; X)}_{\text{minimal sufficient}} \approx \eta' \underbrace{\nabla_\theta \ell(\theta_0; X)}_{\text{near-sufficient?}}$$

Def The score at θ_0 is $S_{\theta_0}(X) = \nabla_\theta \ell(\theta_0; X)$

NB: only a statistic if we specify θ_0 (which nbhd)

Key role in asymptotic statistics

(Rough) Idea: $p_{\theta_0+\eta}(x) = e^{\ell(\theta_0+\eta; x)} \approx e^{\eta' S_{\theta_0}(x) + \overbrace{O(\|\eta\|^2)}^{\text{important term}}} p_{\theta_0}(x)$

Differential Identities (assuming enough regularity)

$$(*) \qquad 1 = \int_{\mathcal{X}} e^{\ell(\theta;x)}\, d\mu(x)$$

$$\frac{\partial}{\partial\theta_j}: \qquad 0 = \int \frac{\partial}{\partial\theta_j}\ell(\theta;x)\, e^{\ell(\theta;x)}\, d\mu(x)$$

$$\Rightarrow \quad \mathbb{E}_\theta\left[\nabla \ell(\theta;X)\right] = 0$$

needs to be same θ

$$\frac{\partial}{\partial\theta_k}: \qquad 0 = \int \left(\frac{\partial^2 \ell}{\partial\theta_j \partial\theta_k} + \frac{\partial \ell}{\partial\theta_j}\cdot\frac{\partial \ell}{\partial\theta_k}\right) e^{\ell}\, d\mu$$

$$= \mathbb{E}_\theta\left[\frac{\partial^2 \ell}{\partial\theta_j \partial\theta_k}\right] + \mathbb{E}_\theta\left[\frac{\partial \ell}{\partial\theta_j}\,\frac{\partial \ell}{\partial\theta_k}\right]$$

$$\Rightarrow \quad \mathrm{Var}_\theta\left[\nabla \ell(\theta;X)\right] = \mathbb{E}_\theta\left[-\nabla^2 \ell(\theta;X)\right]$$

same θ

<u>Def</u> $J(\theta) = \mathrm{Var}_\theta(S_\theta(X))$ called <u>Fisher information</u>

Central quantity in asymptotic statistics

Score in i.i.d. sample

Assume $X_1, \dots, X_n \overset{iid}{\sim} p_\theta^{(1)}(x) \qquad \theta \in \Theta \subseteq \mathbb{R}^d$

p_θ "regular": common support, $\mathbb{E}_\theta \|\nabla \ell(\theta; x)\|^2 < \infty$

$$X \sim p_\theta(x) = \prod_i p_\theta^{(1)}(x_i)$$

Let $\ell_1(\theta; x_i) = \log p_\theta^{(1)}(x_i)$,

$$\ell(\theta; x) = \sum_i \ell_1(\theta; x_i)$$

$$S_\theta(x) = \nabla \ell(\theta; x)$$

$$= \sum_i \nabla \ell_1(\theta; x_i) = \sum_i \overbrace{S_\theta^{(1)}(x_i)}^{\text{univariate score}}$$

$$J(\theta) = \sum_i \mathrm{Var}(S_\theta^{(1)}(x_i)) = n \overbrace{J_1(\theta)}^{\text{univariate Fisher info}}$$

Preview of asymptotics

Note $\frac{1}{n} \nabla^2 \ell(\theta; X) = \frac{1}{n} \sum_i \nabla^2 \ell(\theta; X_i) \xrightarrow{LLN} -J_1(\theta)$

Taylor expansion $(d=1)$

$$\ell(\theta_0 + \eta; X) - \ell(\theta_0; X) \approx \eta S_\theta(X) + \frac{\ddot{\ell}(\theta; \eta)}{2} \eta^2$$

$$\approx \eta S_\theta(X) - \frac{n J_1(\theta)}{2} \cdot \eta^2$$

Cramèr-Rao Lower Bound (aka Info L.B.)

Suppose $\delta(X)$ unbiased for $g(\theta) \in \mathbb{R}$

$$(*) \qquad g(\theta) = \int \delta(x) e^{\ell(\theta;x)} d\mu(x)$$

$$\frac{\partial}{\partial\theta_j}: \quad \frac{\partial}{\partial\theta_j} g(\theta) = \int \delta(x) \cdot \frac{\partial}{\partial\theta_j} \ell(\theta;x) \, e^{\ell(\theta;x)} d\mu(x)$$

$$\nabla g(\theta) = \mathbb{E}_\theta\left[\delta(X) \cdot S_\theta(X)\right]$$

$$= \mathrm{Cov}_\theta\left(\delta(X), S_\theta(X)\right)$$

Leads to variance lower bound:

$$(d=1): \quad \mathrm{Corr}_\theta^2(\delta, S_\theta) = \frac{\mathrm{Cov}_\theta^2(\delta, S_\theta)}{\mathrm{Var}_\theta(\delta)\ \mathrm{Var}_\theta(S_\theta)} = \frac{\dot{g}(\theta)^2}{\mathrm{Var}_\theta(\delta)\, J(\theta)}$$

<u>Rearrange</u>: $\mathrm{Var}_\theta(\delta) = \frac{\dot{g}(\theta)^2}{J(\theta)} \Big/ \underbrace{\mathrm{Corr}_\theta^2(\delta, S_\theta)}_{\leq 1}$

$$\overset{\text{CRLB}}{\geq} \dot{g}(\theta)^2 / J(\theta)$$

<u>Interpretation</u> No unbiased estimator has $\mathrm{Var} < \frac{\dot{g}(\theta)^2}{J(\theta)}$

If $g(\theta) = \theta$, $\mathrm{Var}_\theta(\delta) \geq \frac{1}{J(\theta)}$

$$= \frac{1}{n J_1(\theta)} \quad \text{for i.i.d.} \quad (\text{sd} \asymp n^{-1/2})$$

Multivariate CRLB

For $\theta \in \mathbb{R}^d$, $g(\theta) \in \mathbb{R}$:

$$\text{Var}_\theta(\delta) \geq \nabla g(\theta)' J(\theta)^{-1} \nabla g(\theta)$$

Proof any $a \in \mathbb{R}^d$, $\text{Cov}_\theta(\delta, a' S_\theta) = a' \nabla g(\theta)$

$$\Rightarrow \text{Var}_\theta(\delta) \geq \frac{\text{Cov}_\theta(\delta, a' S_\theta)^2}{\text{Var}_\theta(a' S_\theta)} = \frac{a' \nabla g \nabla g' a}{a' J(\theta) a}$$

Rayleigh quotient, maximized by $a^* = J(\theta)^{-1} \nabla g(\theta)$

$$\rightsquigarrow \text{Var}_\theta(\delta) \geq \nabla g(\theta)' J(\theta)^{-1} \nabla g(\theta) \qquad \boxtimes$$

Score in Exp. Family

Ex $p_\eta(x) = e^{\eta' T(x) - A(\eta)} h(x)$

$$\ell(\eta; x) = \eta' T(x) - A(\eta) + \log h(x)$$

Score:

$$\nabla \ell(\eta; x) = T(x) - \nabla A(\eta)$$

$$\Rightarrow S_\eta(x) = T(x) - \mathbb{E}_\eta T(X)$$

Fisher info (1)

$$\mathrm{Var}_\eta(S_\eta(X)) = \mathrm{Var}_\eta(T(X)) = \nabla^2 A(\eta)$$

Fisher info (2)

$$\nabla^2 \ell(\eta; X) = -\nabla^2 A(\eta)$$

$$\mathbb{E}_\eta\left[-\nabla^2 \ell(\eta; X)\right] = \nabla^2 A(\eta) \quad \checkmark$$

Curved Exp. Family

Ex $p_\theta(x) = e^{\eta(\theta)' T(x) - A(\eta(\theta))} h(x)$ $\quad \theta \in \mathbb{R}$, $\eta(\theta) \in \mathbb{R}^d$

$$\ell(\theta; x) = \eta(\theta)' T(x) - A(\eta(\theta)) + \log h(x)$$

Score:

$$\dot{\ell}(\theta; x) = \dot{\eta}(\theta)' T(x) - \dot{\eta}(\theta)' \nabla A(\eta(\theta))$$

$$\Rightarrow S_\theta(x) = \dot{\eta}(\theta)' \big(\underbrace{T(x) - \mathbb{E}_\theta T(x)}_{\text{score for ambient family}} \big) = \dot{\eta}(\theta)' S^{(\eta)}_{\eta(\theta)}(x)$$

Fisher info (1)

$$\mathrm{Var}_\theta(S_\theta(x)) = \dot{\eta}(\theta)' \mathrm{Var}(S^{(\eta)}_{\eta(\theta)}(x)) \dot{\eta}(\theta)$$

$$= \dot{\eta}(\theta)' J^{(\eta)}(\eta(\theta)) \dot{\eta}(\theta)$$

Fisher info (2)

$$\ddot{\ell}(\theta; x) = \ddot{\eta}(\theta)' (T(x) - \mathbb{E}_\theta T(x)) - \dot{\eta}(\theta)' \nabla^2 A(\eta(\theta)) \dot{\eta}(\theta)$$

$$\mathbb{E}_\theta[-\ddot{\ell}(\theta; x)] = -\, -\, \dot{\eta}(\theta)' \nabla^2 A(\eta(\theta)) \dot{\eta}(\theta)$$

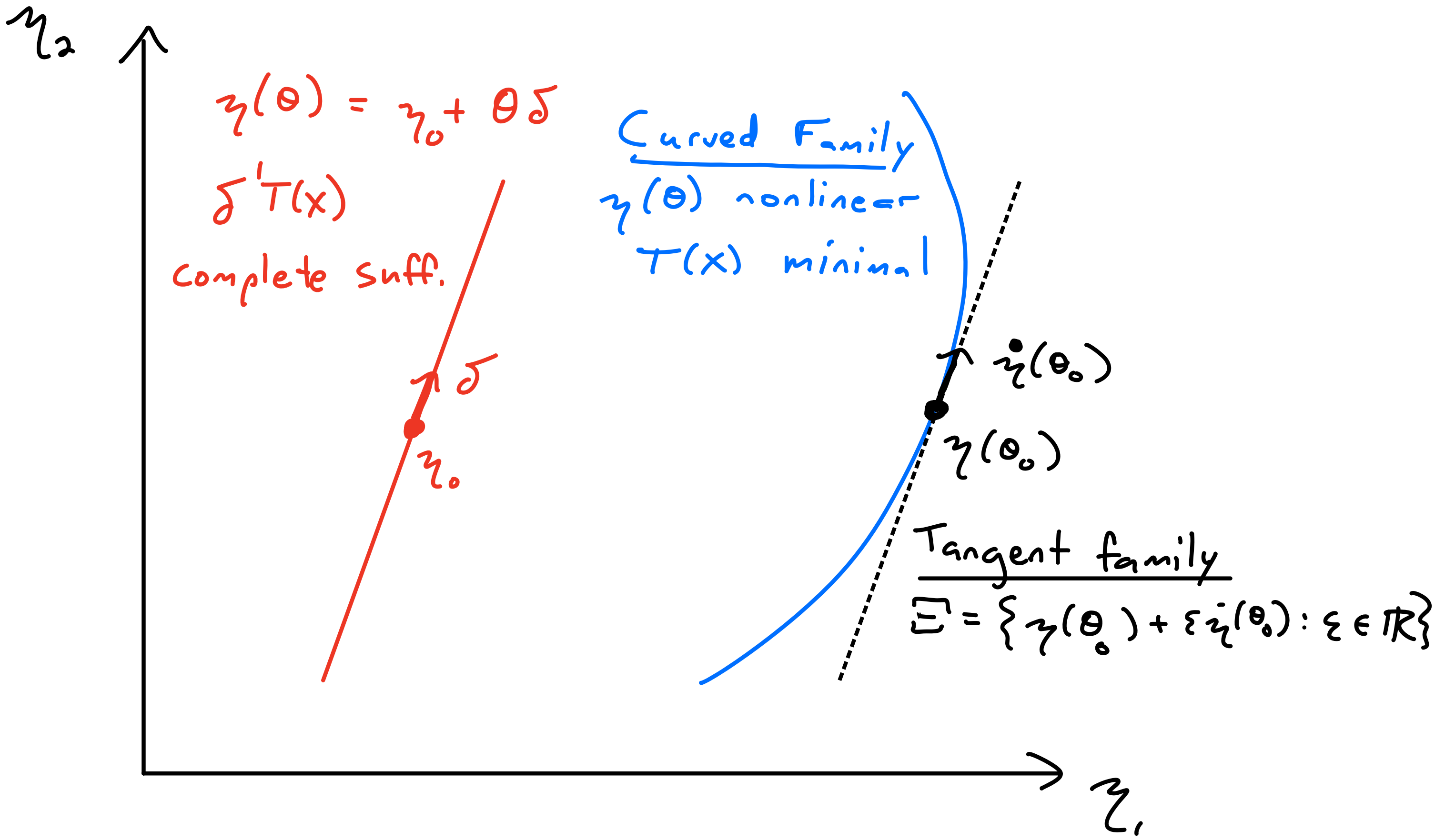
$$= \dot{\eta}(\theta)' J^{(\eta)}(\eta(\theta)) \dot{\eta}(\theta)$$

Motivation: Tangent family

Concrete if $\mathcal{P}$ is curved exp. family

$$p_\theta(x) = e^{\eta(\theta)'T(x) - A(\eta(\theta))} h(x) \qquad \eta: \mathbb{R} \to \mathbb{R}^2$$

$$\Xi = \{\eta(\theta) : \theta \in \mathbb{R}\}$$



$$q_\varepsilon(x) = e^{(\eta(\theta_0) + \varepsilon\dot\eta(\theta_0))'T(x) - A(\cdots)} h(x)$$

$$= e^{\varepsilon \underbrace{\dot\eta(\theta_0)'(T(x) - \mathbb{E}_{\theta_0}T)}_{S_{\theta_0}(x)} - B(\varepsilon)} k(x)$$

Complete sufficient for tangent family at θ_0

Ex Curved Gaussian family

$X_1, \ldots, X_n \overset{iid}{\sim} N_d(\mu(\theta), I_d)$ $\qquad$ $\theta \in \mathbb{R}$, $\mu(\theta) \in \mathbb{R}^d$

$I_d = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix} \in \mathbb{R}^{d \times d}$

Ambient family:

$$S_\mu^{(n)}(x) = \sum_{i=1}^n X_i - n\mu = n(\bar{X} - \mu)$$

$$J^{(n)}(\mu) = \mathrm{Var}_\mu(S_\mu(x)) = n I_d$$

Curved subfamily:

$$S_\theta(\bar{x}) = n\,\dot{\mu}(\theta)'(\bar{x} - \mu(\theta)), \qquad J(\theta) = \underset{\text{sample size}}{n}\ \underbrace{\|\dot{\mu}(\theta)\|^2}_{\text{squared rate of change wrt } \theta}$$

Ex: $d = 2$, $\mu(\theta) = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$ $\qquad \theta \in [0, \pi/2]$